EXPERIMENT 6

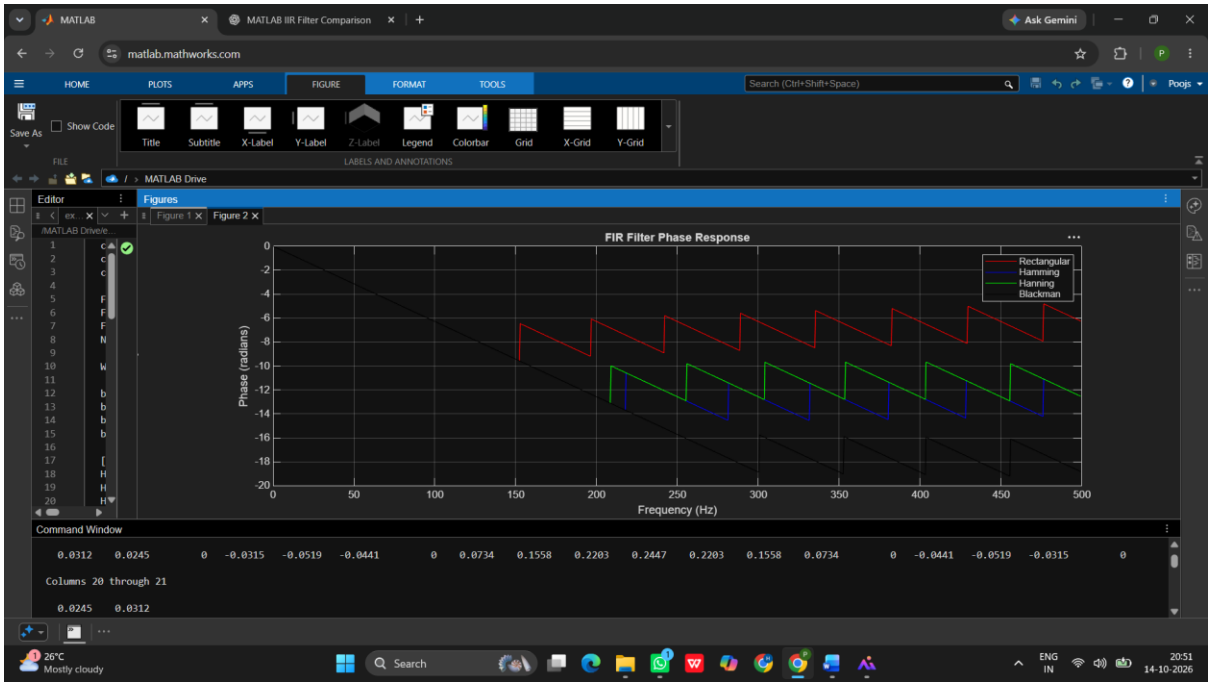
NAME: Pooja V

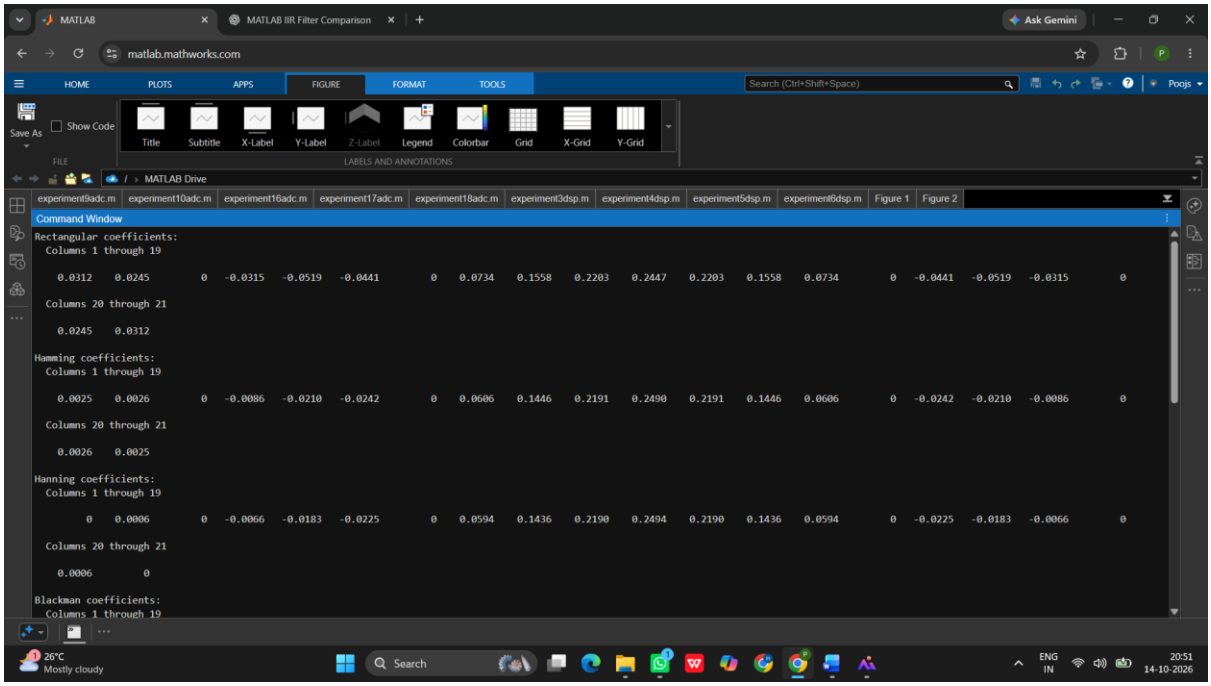
REG NO: 192512463

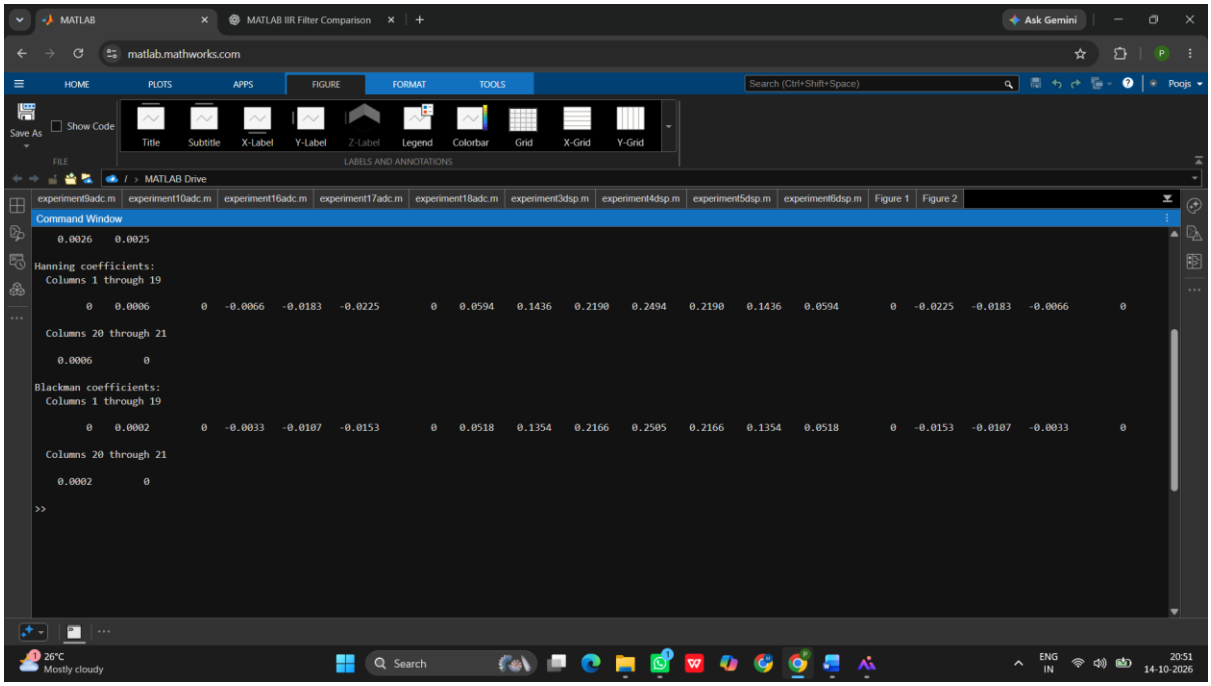
PROGRAM:

```
clc;
clear;
close all;
Fs = 1000;
Fp = 100;
Fst = 150;
N = 20;
Wc = (Fp + Fst)/2/(Fs/2);
...legend('Rectangular','Hamming','Hanning','Blackman');
disp('Rectangular coefficients:');
disp(b1);
disp('Hamming coefficients:');
disp(b2);
disp('Hanning coefficients:');
disp(b3);
disp('Blackman coefficients:');
disp(b4);
```


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FIR Filter Window Comparison
Rectangular
Hamming
Hanning
Blackman
Magnitude (dB)
Frequency (Hz)
Command Window
0.0312 0.0245 0 -0.0315 -0.0519 -0.0441 0 0.0734 0.1558 0.2203 0.2447 0.2203 0.1558 0.0734 0 -0.0441 -0.0519 -0.0315 0
Columns 20 through 21
0.0245 0.0312
26°C
Mostly cloudy
Search
ENG
IN
20:50
14-10-2026


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FIR Filter Phase Response
Rectangular
Hamming
Hanning
Blackman
Phase (radians)
Frequency (Hz)
Command Window
0.0312 0.0245 0 -0.0315 -0.0519 -0.0441 0 0.0734 0.1558 0.2203 0.2447 0.2203 0.1558 0.0734 0 -0.0441 -0.0519 -0.0315 0
Columns 20 through 21
0.0245 0.0312
26°C
Mostly cloudy
Search
ENG
IN
20:51
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experiment10adc.m
experiment16adc.m
experiment17adc.m
experiment18adc.m
experiment3dsp.m
experiment4dsp.m
experiment5dsp.m
experiment6dsp.m
Figure 1
Figure 2
Command Window
Rectangular coefficients:
Columns 1 through 19
0.0312 0.0245 0 -0.0315 -0.0519 -0.0441 0 0.0734 0.1558 0.2203 0.2447 0.2203 0.1558 0.0734 0 -0.0441 -0.0519 -0.0315 0
Columns 20 through 21
0.0245 0.0312
Hamming coefficients:
Columns 1 through 19
0.0025 0.0026 0 -0.0086 -0.0210 -0.0242 0 0.0606 0.1446 0.2191 0.2490 0.2191 0.1446 0.0606 0 -0.0242 -0.0210 -0.0086 0
Columns 20 through 21
0.0026 0.0025
Hanning coefficients:
Columns 1 through 19
0 0.0006 0 -0.0066 -0.0183 -0.0225 0 0.0594 0.1436 0.2190 0.2494 0.2190 0.1436 0.0594 0 -0.0225 -0.0183 -0.0066 0
Columns 20 through 21
0.0006 0
Blackman coefficients:
Columns 1 through 19
26°C
Mostly cloudy
Search
ENG
IN
20:51
14-10-2026


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experiment3dsp.m
experiment4dsp.m
experiment5dsp.m
experiment6dsp.m
Figure 1
Figure 2
Command Window
0.0026 0.0025
Hanning coefficients:
Columns 1 through 19
0 0.0006 0 -0.0066 -0.0183 -0.0225 0 0.0594 0.1436 0.2190 0.2494 0.2190 0.1436 0.0594 0 -0.0225 -0.0183 -0.0066 0
Columns 20 through 21
0.0006 0
Blackman coefficients:
Columns 1 through 19
0 0.0002 0 -0.0033 -0.0107 -0.0153 0 0.0518 0.1354 0.2166 0.2505 0.2166 0.1354 0.0518 0 -0.0153 -0.0107 -0.0033 0
Columns 20 through 21
0.0002 0
>>
26°C
Mostly cloudy
Search
ENG
IN
20:51
14-10-2026